

Figure 4: Advanced BOT configuration



Figure 5: Creating groups

like 'MailAccess PCs can use MailingServices' and click **Next**. Here, you can review the rule. Click **Save** to create the rule.

Internet access is to be allowed only through the IPCop Proxy. Create an access rule for the WebAccess PCs, to permit access to the IPCop Proxy service:

- Source—Custom addresses—WebAccess
 - Destination—IPCop Access—User Services—Custom services: IPCopProxy
 - Rule Enabled
 - Remark—WebAccess PCs can access IPCop Proxy
- Save the rule.

All BOT rules are available for viewing, editing and duplicating from the **Firewall—Block Outgoing Traffic** link. This completes BOT configuration. The access controls in Table 2 are implemented. WebAccess PCs can browse the Internet, but cannot access emails via email clients, while MailAccess PCs can use mail clients but cannot browse the Internet.

Configuration verification

Try browsing the Web from a PC with an address not mentioned in the WebAccess range, say 192.168.51.25. The browsing does not work, and the log confirms this by showing **GREEN DROP** under the **Chain** column. The log shows **Dst Port** as 800, by its default service, **MDBS_DAEMON**. Now, try browsing the Internet from a PC with an address in the WebAccess range, say 192.168.51.35—and as expected, it works. You can also check **Logs—Proxy Logs** in the IPCop Web GUI to

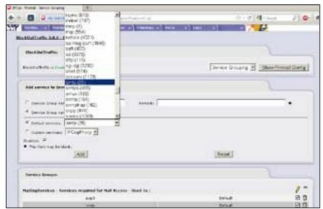


Figure 6: Creating the MailingServices group



Figure 7: Firewall Log denying Internet Access

confirm this. You can further check POP and SMTP access from email clients on computers in the two groups, and see if it works as desired.

Troubleshooting

A few services may stop functioning after enabling BOT. This is because BOT blocks all traffic that is not explicitly allowed. Go to your BOT settings and enable the logging of packets that have not matched a BOT rule. Re-enable BOT and try your service again. Then look into the IPCop firewall logs under **Logs—Firewall Log** for the blocked traffic—to get hints about which rule is missing. Now you should be able to create the missing BOT **Accept** rule. **END**

References

- [1] IPCop website: www.ipcop.org
- [2] BOT FAQs: <http://blockouttraffic.de/faq.php>
- [3] IPCop support community support forum: www.ipcoops.com

By: Rajesh Deodhar

The author is a BE in Industrial Electronics, a CISA (Certified Information Systems Auditor), holds a DCL (Diploma in Cyber Law) and is a CCNA (Certified Cisco Certified Network Associate). He has more than 15 years of experience in the field of computer hardware, networking, firewalls, and IS auditing as a director at Omega Systems and Services Pvt Ltd, Pune. He has deployed more than 50 IPCop-based firewalls in various verticals using different configurations. Please send your feedback/queries to rajesh@omegasystems.co.in.